INCH-POUND

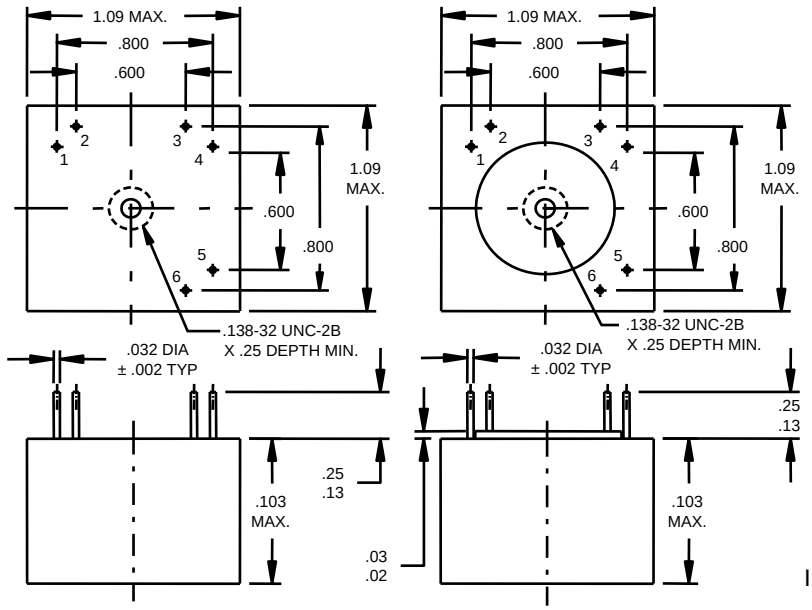
MIL-PRF-27/42C
23 September 2009
SUPERSEDING
MIL-PRF-27/42B
7 February 1984

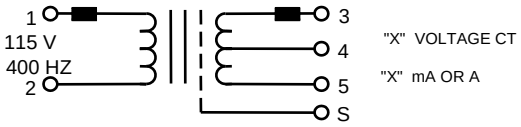
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 6 VOLT AMPERES, 400 HZ, ELECTROSTATIC SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification sheet and MIL-PRF-27.





MAX W.V. , "X"V"
MAX ALT 70 .000

(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING

Inches	mm
.002	0.05
.02	0.5
.03	0.8
.032	0.81
.13	3.3
.138	.51
.25	6.4
.600	15.24
.75	19.1
.800	20.32
1.09	27.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations

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REQUIREMENTS: (When numbers in parentheses, such as (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 2.2 ounces, maximum.

Duty cycle: Continuous.

Case: Plastic encapsulated.

Material: Plastic.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55˚C to +130˚C, maximum.

Terminals: Printed circuit pins - tinned brass.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts rms, 400 Hz across (1-2): Current in (1-2) shall not exceed 9 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a restrictive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 80 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{\text{Voltage (no Load) - Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

shall not exceed 10 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 25°C, maximum.

Vibration: MIL-STD-202, method 204, test condition D.

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Marking location: See figure 1.

VERIFICATION

Extent of qualification

Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-PRF-27/37 through MIL-PRF-27/47 inclusive, all parts.

Qualification testing and approval to M27/42-44 shall be sufficient to grant qualification approval to M27/42-01 through M27/42-44.

Part or Identification Number (PIN): M27/42-01 A or B.

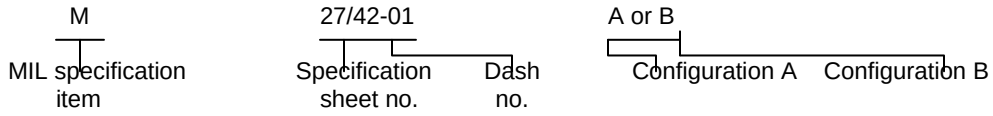

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TABLE I. Electrical ratings.

Dash No.	Type designation	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms) (MAX)
01 A or B	620	6	1.100	325
02 A or B	621	6.3	0.953	325
03 A or B	622	7	0.857	325
04 A or B	623	7.5	0.800	325
05 A or B	624	8	0.750	325
06 A or B	625	9	0.667	325
07 A or B	626	10	0.600	325
08 A or B	627	11	0.545	325
09 A or B	628	12	0.500	325
10 A or B	629	12.6	0.476	325
11 A or B	630	13	0.462	325
12 A or B	631	15	0.400	325
13 A or B	632	16	0.375	325
14 A or B	633	18	0.333	325
15 A or B	634	20	0.300	325
16 A or B	635	22	0.272	325
17 A or B	636	24	0.250	325
18 A or B	637	26	0.230	325
19 A or B	638	28	0.214	325
20 A or B	639	30	0.200	325
21 A or B	640	33	0.182	325
22 A or B	641	36	0.167	325
23 A or B	642	39	0.154	325
24 A or B	643	43	0.140	325
25 A or B	644	47	0.128	325
26 A or B	645	51	0.118	325
27 A or B	646	56	0.107	325
28 A or B	647	62	0.097	325
29 A or B	648	68	0.088	325
30 A or B	649	75	0.080	325
31 A or B	650	82	0.073	325
32 A or B	651	91	0.066	325
33 A or B	652	100	0.060	325
34 A or B	653	110	0.055	325
35 A or B	654	115	0.052	325
36 A or B	655	120	0.050	325
37 A or B	656	130	0.046	325
38 A or B	657	150	0.040	325
39 A or B	658	180	0.033	325
40 A or B	659	200	0.030	440
41 A or B	660	220	0.027	440
42 A or B	661	240	0.025	440
43 A or B	662	270	0.022	580
44 A or B	663	300	0.020	580

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-PRF-27/37	MIL-PRF-27/38	MIL-PRF-27/39	MIL-PRF-27/40
MIL-PRF-27/41	MIL-PRF-27/43	MIL-PRF-27/44	MIL-PRF-27/45
MIL-PRF-27/46	MIL-PRF-27/47	MIL-STD-202	

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2009-041)

Review activities:
Army - CR4, MI
Navy - AS, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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